A ticking time bomb: Bioinformatics analysis on environmental *Serratia* species to assess the emergence of antibiotic and virulence traits

Adi Chauhan and Aryan Taywade

Summary

Exposure of environmental bacteria to long-term pollution can facilitate recruitment of multidrug-resistant (MDR) genes; these bacteria are referred to as "superbugs", leading to potential pathogenic outbreaks. In fact, antimicrobial resistance is projected to cause approximately 10 million deaths annually by year 2050, outpacing cancer- the current leading cause of human mortality. Moreover, at least 50% of the deaths reported from the ongoing COVID-19 pandemic were due to secondary bacterial infections and not *per-se* by the COVID-19 pathogen. To assess the development of superbugs in the environment, we worked on a bacterial genus- *Serratia* species, which are either classified as the medical type (pathogenic) or environmental type. A central question in our research was: are environmental *Serratia* acquiring virulence and MDR abilities such that these can cause future outbreaks like the COVID-19 pandemic? To address this overarching question, we used bioinformatics to analyze a recently reported *Serratia marcescens* species strain SRS-8-S-2018 as a model organism, shown to resistant heavy metals and multiple antibiotics. Of major note was the presence of specialty genes for antibiotic resistance (63), virulence factors (93), drug targets (165), and transporters (261)- all of which can render MDR capabilities in this strain. Finally, comparison of our strain, which is an environmental type, revealed closer affiliation with the medical type *Serratia* species, confirming the MDR evolution of environmental *Serratia,* unequivocally showing that long term exposure of contaminants are driving MDR. Overall, such studies will eventually identify appropriate mitigation strategies to combat the global spread of superbugs.

Introduction

https://www.postandcourier.com/news/deadly-legacy-savannah-river-site-near-aiken-one-of-the-most-contaminated-places-on-earth/article_c525f484-12ff-11e7-9579-f0d72fccae53.html

Savannah River Site (SRS) is an approximately 800-km^2 former nuclear weapons production facility located along the Savannah River near Aiken, SC.

The site is co-contaminated by heavy metals and radionuclides and the strain in this study was isolated from this site.

- The bacterial genus *Serratia* is known to be pathogenic (medical-type) or can also be found living in the environment and perform symbiotic roles with other organisms.
- Due to stress posed by environmental pollutants, *Serratia* have developed resistances against environmental pollutants, such as heavy metals, as well as multiple antibiotics, referred to as multiple drug resistance (MDR) abilities.
- **A central question in our research is as follows: Are environmental *Serratia* acquiring virulence and MDR abilities such that these organisms can cause bacterial outbreaks similar to the COVID-19 in the future?**
- To address the overarching question above, we utilized bioinformatics analysis on a recently reported Serratia marcescens species strain SRS-8-S-2018, shown to be resistant to heavy metals such as mercury and uranium as well as possessing MDR abilities (Gendy et al., 2020).
- However, no information exists on how this strain has become MDR. This genome-enabled study provides a better understanding as to how the *Serratia* species acquires pathogenicity when exposed to certain contaminants.
- Overall, our goal was to study *Serratia* spp. as a model organism to study MDR and such that drug and pollutant resistance can be mitigated for environmental stewardship.

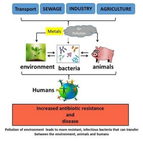


Image taken from https://le.ac.uk/lemid/strategic-areas/air-pollution

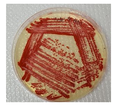


Image taken from Jaswal et. al., Unpublished work

Overarching Hypotheses

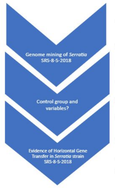
If environmental microorganisms, such as *Serratia* spp., are exposed to long-term stress posed by pollutants, such as heavy metals, then they develop resistances to antibiotics and pathogenicity due to recruitment of virulence genes.

Objectives

Our main objectives are as follows:

1) Study the development of bacterial resistance to environmental pollutants and antibiotics using *Serratia* spp. as the model organism. *Serratia* species are opportunistic pathogens and hence serve as a good model to understand the development of MDR in the environment.
2) Initiate mitigation and stewardship efforts as pertaining to judicious usage of antibiotics for humans, livestock, poultry and other uses.
3) Study the development of bacterial resistance to environmental pollutants and antibiotics using Serratia spp. as the model organism. Serratia species are opportunistic pathogens and hence serve as a good model to understand the development of MDR in the environment.

Materials and Methods



1. Whole genome sequence of *Serratia* strain SRS-8-S-2018, as the model organism isolated from a long-term contaminated site was chosen for this study. We found the whole genome sequence (WGS) of this strain reported in the literature (Gendy, S et al., 2020), which was downloaded from the NCBI genome repository (Tatusova, T et al., 2016).
2. For the bioinformatics analysis, control group represented the *Serratia* species SRS-8-S-2018 and variable groups were represented by other *Serratia* strains for which WGS are available in NCBI genome repository (Tatusova, T et al., 2016)
3. Phylogenomic affiliation for strain SRS-8-S-2018 relative to other strains available in databases was conducted using the Onecodex pipeline (Minot, S et al., 2015).
4. In this regard, our control group was *Serratia* strain SRS-8-S-2018 and variable groups were the cohort of other Serratia strains against which this comparative genomics project was run.
5. A genomic map of *Serratia* strain SRS-8-S-2018 was created using the PATRIC pipeline. This facilitated identification of genomic features that further led us to identify how many drug resistance and virulence genes our strain possessed.
6. The next step in the bioinformatics analysis is to annotate the genome sequence, which is a series of steps to identify locations of genes within an organism and to predict the genomic functions. Annotation was performed using the PATRIC (Pathosystems Resource Integration Center) annotation is based on the locations of the gene that is found
7. Further gene prediction was performed using additional pipelines such as RAST(Aziz, R. K. et al., 2008) and IMG/er pipelines (chen, I.-M. A. et al., 2018).
8. Presence of speciality genes for virulence, pathogenicity and drug resistance were studied.
9. One mechanism by which bacteria acquire genes for adaptation and to survive in the presence of contaminants is via horizontal gene transfer events, which leaves genomic segments within the bacterial genome called genomic islands (GEIs). Genomic islands present in *Serratia* species SRS-8-S-2018 were identified by different bioinformatic approaches such as island viewer (Bertelli, C et al., 2017)
10. Statistical analysis was conducted between the control (*Serratia* species SRS-8-S-2018) and 14 closest Serratia strains (variable groups). Those *Serratia* species were selected that showed closer affiliation with both the medical types relative to the environmental types.

Results and Discussion

Experiment #1: Comparison of Phylogenetic analysis of strain SRS-8-S-2018 using the OneCodex Bioinformatic pipeline. We found that our strain was most closely related to Serratia marcescens WW4 with 35 reads, which accounts for 22% of all reads.

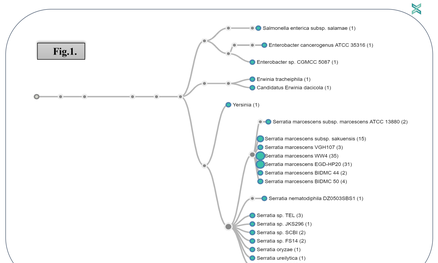


Figure 1 produced by Adi Chauhan and Aryan Taywade using the One Codex - microbial bioinformatics pipeline, https://www.onecodex.com/

Fig. 1. Genome-wide phylogenetic assessment of strain SRS-8-S-2018 provided evidence for affiliation with *Serratia marcescens*. The figure above was obtained using default settings in the One Codex workflow, which identifies microbial sequences using a "k-mer based" taxonomic classification algorithm through a web-based data platform, using a reference database that currently includes approximately 40,000 bacterial, viral, fungal, and protozoan genomes.

Results and Discussion

Experiment #2: Genomic mapping through the PATRIC Pipeline to discover the genomic features of the strain.

Figures 2, 3, 4 produced by Adi Chauhan and Aryan Taywade using PATRIC (Pathosystems Resource Integration Center), https://www.patricbrc.org/

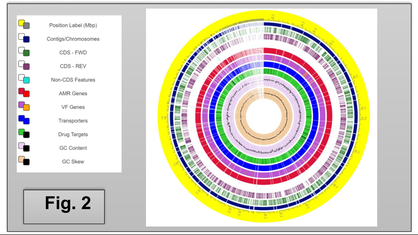


Fig. 2

Fig. 2. Genomic mapping of *Serratia* sp. strain SRS-8-S-2018. Shown is a circular genome map of strain SRS-8-S-2018, obtained using the default settings within the PATRIC pipeline. The map consists of 7 circles from inside out; first Position Label (Mbp), the second contain AMR Genes, the third contains VF Genes, the fourth contains Transporters, the fifth contains Drug Targets, the sixth contains GC Content, and the seventh contains GC Skew.

Genomic Features		Sequence Info		Protein Features	
CDS	5352	Genome Size/Length	5323630	Hypothetical Proteins	930
tRNA	56	Contigs	159	Protein with Functional Assignments	4422
rRNA	1	GC Content	59.480824	Proteins with EC Number Assignments	1335
				Proteins with GO Assignments	1102
				Proteins with Pathway Assignments	953
				Proteins with Subsystem Assignments	1783

Table 1. Genomic features of strain SRS-8-S-2018. Categories marked with the red arrows represent those that render functional traits that potentially facilitate the strain's survival in a heavy metal and radionuclide contaminated soil habitat.

Experiment #3: Identify the suite of specialty genes, such as virulence and antibiotic resistance in strain SRS-8-S-2018 using in-silico analysis.

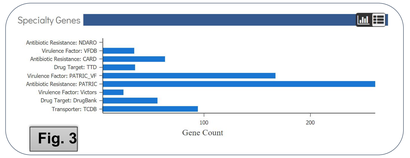


Fig. 3

Fig. 3. Shown are total speciality genes identified in *Serratia* sp. strain SRS-8-S-2018 using PATRIC workflow.

Figures 2, 3, 4 produced by Adi Chauhan and Aryan Taywade using PATRIC (Pathosystems Resource Integration Center), https://www.patricbrc.org/

Experiment #4: Using PATRIC, we identified the subsystems in the strain in order to understand the distribution of genes within super classes.

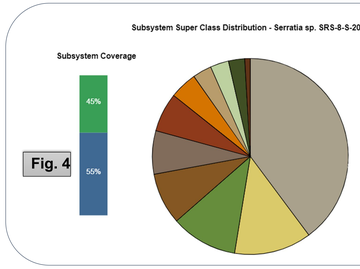


Fig. 4

Fig. 4. Subsystems identified in strain *Serratia* sp. strain SRS-8-S-2018. Subsystems are curated sets of biologically related functional roles across a reference set of genomes. in PATRIC, there are 11 total subsystems as shown above. Notaly, the virulence function was identified as the fourth largest subsystem identified in this strain with a total of 207 genes.

Experiment #5: Identifying genomic islands of the strain to find what genes have been transferred as a result of horizontal transport.

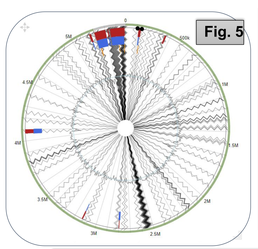


Fig. 5. Shown are the putative genomic islands (GEIs) of SRS-8-S-2018 strain. The outer green circle represents the scale line in Mbps and GEIs obtained from each of the following methods are shown in color: SIGI-HMM (orange), IslandPath-DIMOB (blue), and integrated detection (red), respectively. A total of 26 GEIs were identified in this strain with many genes encoded on these GEIs with functions related to transport, efflux, transposase and phage elements.
One of the mechanisms for GEIs to be present in bacterial genomes is via bacteriophage mediated transfer. Note that viruses that attack bacteria (bacteriophages) can leave their signature genetic elements within their bacterial host genomes; higher phage remnants indicate recruitment of genes via virus mediated attacks during the evolution of bacterial strain. Some such traits include virulence factors, antibiotic resistance, and GEIs.

Figure 5 produced by Adi Chauhan and Aryan Taywade using Islandviewer 4 - computation tool for genomic island prediction, http://www.pathogenomics.sfu.ca/islandviewer/

Experiment #6: Comparison of *Serratia* sp. SRS-8-S-2018 (environmental type) with 14 closest *Serratia* strains that are environmental and medical types using comparative genomics workflows.

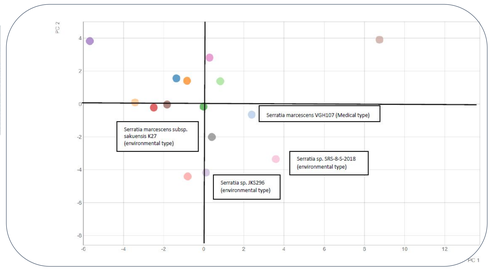


Fig. 6. Principal components analysis (PCA) of *Serratia* sp. SRS-8-S-2018 with 14 closest *Serratia* strains showed closer affiliation with medical type relative to the environmental types.

Figure 6 produced by Adi Chauhan and Aryan Taywade using IMG (The Integrated Microbial Genomes system) - genome browsing and annotation, https://img.jgi.doe.gov/

Conclusions

Overall, the genomic features of strain SRS-8-S-2018 was as follows: genome size was 5,323,630 bp, with 5,352 coding DNA sequences, and an average G + C content of 59.48.

Of major note was the presence of specialty genes for antibiotic resistance (63), virulence factors (93), drug targets (165), and transporters (261)- all of which can render MDR capabilities to this strain. Using the PATRIC database we found 11 total subsystems, and the virulence function was identified as the fourth largest subsystem identified in this strain with a total of 207 genes, showing the capabilities of virulence in this strain.

Using the PATRIC bioinformatics analysis, we identified certain genomic features of the strain through a circular genome map of Strain SRS-8-S-2018. Virulence factors (VF) and MDR Genes were found in abundance.

Using the Island Viewer, we were able to identify the genomic islands (GEIs) of the strain to find what genes have been transferred as a result of potential horizontal transport to this strain. A total of 26 GEIs were identified in this strain with many genes encoded on those GEIs with functions related to transport, efflux, transposase, and phage elements.

Finally, in the comparison of our strain, an environmental type, revealed its closest affiliation with medical type *Serratia* species, which confirms that at a functional level the environmental *Serratia* are similar to their medical counterparts, showing that long term exposure of environmental contaminants are likely driving MDRs in the environment.

Citations

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